

Standard Operating Procedure (SOP): Agricultural Drone Spraying

ICAR & Directorate of Plant Protection, Quarantine & Storage

1. Field Execution Protocol

Drone spraying achieves high precision micro-droplet deposition. Pilots must maintain an altitude of 1.5 to 2.0 meters above crop canopy with a swath width of 3.5 to 4.5 meters.

2. Efficiency Comparison: Manual vs Drone Spraying

Metric	Manual Labor Spraying	Drone Spraying	Improvement
Water Volume	200 - 250 Liters / Acre	10 - 12 Liters / Acre	95% Reduction
Chemical Savings	100% Standard Dosage	75% Ultra-Low Volume	25% Savings
Operational Speed	3.5 Hours / Acre	7 - 10 Minutes / Acre	25x Faster
Labor Health Risk	High Dermal Exposure	Zero Direct Contact	100% Safe

3. Weather Limitations & Safety Limits

Do not operate drones when wind speed exceeds 25 km/h, ambient temperature exceeds 40 degrees C, or during heavy precipitation/fog.